§4 多元复合函数的求导法则

- 链式求导法则
- 全微分形式不变性

复习：

一元复合函数 $y = f(u),\ u = \varphi(x)$

求导法则 $$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x}$$

微分法则及一元函数微分的形式不变性

$$\mathrm{d}y = f'(u)\,\mathrm{d}u = f'(u)\varphi'(x)\,\mathrm{d}x$$

一、链式法则

定理 如果函数$u=\phi(t)$及$v=\psi(t)$都在点 t可导，函数$z=f(u,v)$在对应点(u,v)具有连续偏导数，则复合函数$z=f[\phi(t),\psi(t)]$在对应点t可导，且其导数可用下列公式计算：

$$\frac{dz}{dt}=\frac{\partial z}{\partial u}\frac{du}{dt}+\frac{\partial z}{\partial v}\frac{dv}{dt}.$$

证 设 t 获得增量 Δt,

则 $\Delta u=\phi(t+\Delta t)-\phi(t)$, $\Delta v=\psi(t+\Delta t)-\psi(t)$;

由于函数$z=f(u,v)$在点(u,v)有连续偏导数

$$\Delta z=\frac{\partial z}{\partial u}\Delta u+\frac{\partial z}{\partial v}\Delta v+\varepsilon_1\Delta u+\varepsilon_2\Delta v,$$

当$\Delta u\to 0$，$\Delta v\to 0$时，　$\varepsilon_1\to 0$，$\varepsilon_2\to 0$

$$\frac{\Delta z}{\Delta t}=\frac{\partial z}{\partial u}\cdot\frac{\Delta u}{\Delta t}+\frac{\partial z}{\partial v}\cdot\frac{\Delta v}{\Delta t}+\varepsilon_1\frac{\Delta u}{\Delta t}+\varepsilon_2\frac{\Delta v}{\Delta t}$$

当$\Delta t\to 0$时，　$\Delta u\to 0$，$\Delta v\to 0$

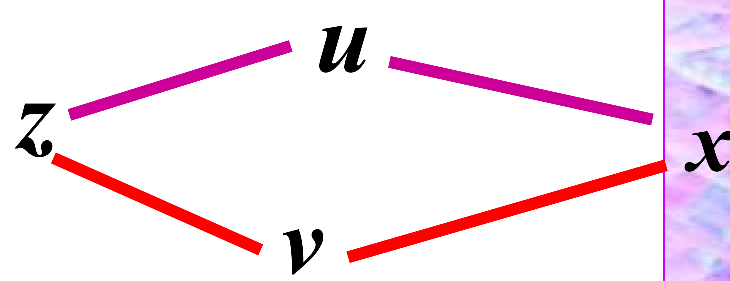
$$\frac{\Delta u}{\Delta t}\to\frac{du}{dt},\qquad \frac{\Delta v}{\Delta t}\to\frac{dv}{dt},$$

$$\frac{dz}{dt}=\lim_{\Delta t\to 0}\frac{\Delta z}{\Delta t}=\frac{\partial z}{\partial u}\cdot\frac{du}{dt}+\frac{\partial z}{\partial v}\cdot\frac{dv}{dt}.$$

例1 设$z=e^{u-2v},u=\sin x,v=e^{x}$ 求：$\frac{dz}{dx}$

解

$$\frac{\partial z}{\partial u}=e^{u-2v}\qquad \frac{\partial z}{\partial v}=-2e^{u-2v}$$



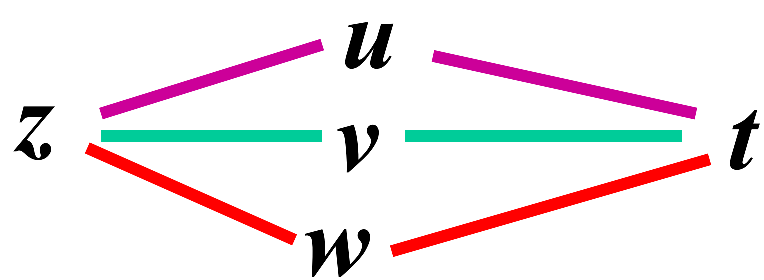
$$\frac{du}{dx}=\cos x\qquad \frac{dv}{dx}=e^{x}$$

$$\frac{dz}{dx}=\frac{\partial z}{\partial u}\frac{du}{dx}+\frac{\partial z}{\partial v}\frac{dv}{dx}=e^{u-2v}(\cos x-2e^{x})$$

推广1：中间变量多于两个的情况.

$$z=f(u,v,w),\quad u=u(t),\quad v=v(t),\quad w=w(t)$$

$$\frac{dz}{dt}=\frac{\partial z}{\partial u}\frac{du}{dt}+\frac{\partial z}{\partial v}\frac{dv}{dt}+\frac{\partial z}{\partial w}\frac{dw}{dt}$$



以上公式中的导数 $\frac{dz}{dt}$ 称为**全导数.**

例2. 设 $z = uv + \sin t$, $u = e^t$, $v = \cos t$, 求全导数 $\dfrac{\mathrm{d}z}{\mathrm{d}t}$.

解:

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\partial z}{\partial u}\cdot\frac{\mathrm{d}u}{\mathrm{d}t} + \frac{\partial z}{\partial v}\cdot\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{\partial z}{\partial t}$$

$$= v\,e^t - u\sin t + \cos t$$

$$= e^t(\cos t - \sin t) + \cos t$$

推广2：中间变量是二元函数的情况

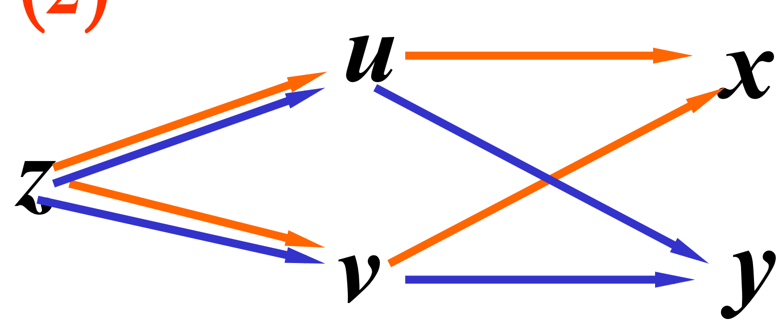
设 $u=\varphi(x,y)$, $v=\psi(x,y)$ 在点 (x,y) 偏导数存在，

$z=f(u,v)$ 在对应点 (u,v) 有连续的偏导数 $\Rightarrow$

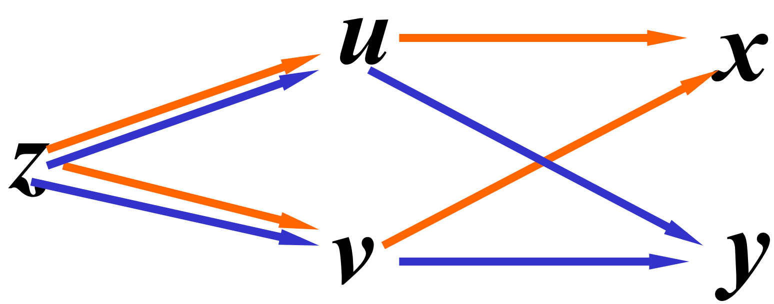
复合函数 $z=f[\varphi(x,y),\psi(x,y)]$ 在点 (x,y) 偏导数存在

且
$$\begin{cases} \dfrac{\partial z}{\partial x}=\dfrac{\partial z}{\partial u}\dfrac{\partial u}{\partial x}+\dfrac{\partial z}{\partial v}\dfrac{\partial v}{\partial x} & (1) \\[2ex] \dfrac{\partial z}{\partial y}=\dfrac{\partial z}{\partial u}\dfrac{\partial u}{\partial y}+\dfrac{\partial z}{\partial v}\dfrac{\partial v}{\partial y} & (2) \end{cases}$$

称为链式法则



链式法则如图示



$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}=f_1'\varphi_1'+f_2'\psi_1'$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}=f_1'\varphi_2'+f_2'\psi_2'$$

证明(仅证第(1)式) $\because$ 求 $\dfrac{\partial z}{\partial x}$, y看作常量.

当x变到$x+\Delta x$, $\Rightarrow$ u变到$u+\Delta_x u$, v变到$v+\Delta_x v$

$\Rightarrow$ z变到$z+\Delta z$

$\because z=f(u,v)$在(u,v)可微, $\quad \rho=\sqrt{(\Delta_x u)^2+(\Delta_x v)^2}$

$$\therefore \Delta z = f(u+\Delta_x u, v+\Delta_x v) - f(u,v)$$

$$=\frac{\partial z}{\partial u}\Delta_x u+\frac{\partial z}{\partial v}\Delta_x v+o(\rho) \quad (\rho\to 0)$$

$$\Delta_x u = u(x+\Delta x, y)-u(x,y)$$

$$\Delta_x v = v(x+\Delta x, y)-v(x,y)$$

两边同除以Δx得

$$\frac{\Delta z}{\Delta x}=\frac{\partial z}{\partial u}\frac{\Delta_x u}{\Delta x}+\frac{\partial z}{\partial v}\frac{\Delta_x v}{\Delta x}+\frac{o(\rho)}{\Delta x}$$

当$\Delta x\to 0$时，有$\Delta_x u\to 0, \Delta_x v\to 0$

$$\therefore \lim_{\Delta x\to 0}\frac{\Delta z}{\Delta x}=\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial x}$$

同理可证 $\frac{\partial z}{\partial y}$

例3 设$z=e^u\sin v$，而$u=xy$，$v=x+y$，

求 $\dfrac{\partial z}{\partial x}$和$\dfrac{\partial z}{\partial y}$.

解 $$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$=e^u\sin v\cdot y+e^u\cos v\cdot 1=e^u(y\sin v+\cos v),$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}$$

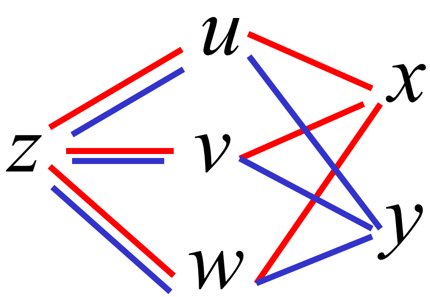
$$=e^u\sin v\cdot x+e^u\cos v\cdot 1=e^u(x\sin v+\cos v).$$

推广3：关于中间变量是二元以上函数的情况

设 $u=\phi(x,y)$、$v=\psi(x,y)$、$w=w(x,y)$ 都在点 (x,y) 具有对 x 和 y 的偏导数，复合函数 $z=f[\phi(x,y),\psi(x,y),w(x,y)]$ 在对应点 (x,y) 的两个偏导数存在，且可用下列公式计算

$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial x}+\frac{\partial z}{\partial w}\frac{\partial w}{\partial x},$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial y}+\frac{\partial z}{\partial w}\frac{\partial w}{\partial y}.$$



特殊情况4 $z=f(u,x,y)$ 其中 $u=\phi(x,y)$

即 $z=f[\phi(x,y),x,y]$, 令 $v=x$, $w=y$,

$$\frac{\partial v}{\partial x}=1,\quad \frac{\partial w}{\partial x}=0,\quad \frac{\partial v}{\partial y}=0,\quad \frac{\partial w}{\partial y}=1.$$

$$\boxed{\frac{\partial z}{\partial x}}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\boxed{\frac{\partial f}{\partial x}},\qquad \boxed{\frac{\partial z}{\partial y}}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y}+\boxed{\frac{\partial f}{\partial y}}.$$

区别类似

两者的区别

把复合函数 $z=f[\phi(x,y),x,y]$ 中的 y 看作不变而对 x 的偏导数

把 $z=f(u,x,y)$ 中的 u 及 y 看作不变而对 x 的偏导数

例4 设$z=f(u,x,y)=6u^2+x^2+4y^3, u=xy$,求$\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$.

解 $\boxed{\dfrac{\partial z}{\partial x}}=\dfrac{\partial f}{\partial u}\cdot\dfrac{\partial u}{\partial x}+\boxed{\dfrac{\partial f}{\partial x}},$

两者的区别

把复合函数$z=f[\phi(x,y),x,y]$中的y看作不变而对x的偏导数

把 $z=f(u,x,y)$ 中的u及y看作不变而对x的偏导数

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial x}\cdot 1=12u\cdot y+2x=12xy^2+2x$$

$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial f}{\partial y}\cdot 1=12u\cdot x+12y^2=12x^2y+12y^2$$

分析给出公式的规律如下：

(1)偏导公式的个数与自变 量的个数相同．

(2)每个偏导公式的项数是 通向自变量途径的数目.

(3)每项结构，“$\dfrac{\partial 因}{\partial 中}\dfrac{\partial 中}{\partial 自}$”--与一元相同．

求导过程中要注意以下问题：

(1) 画变量关系图，分析哪些是中间变量，哪些是自变量。

(2)遵照上面的规律求偏导 数

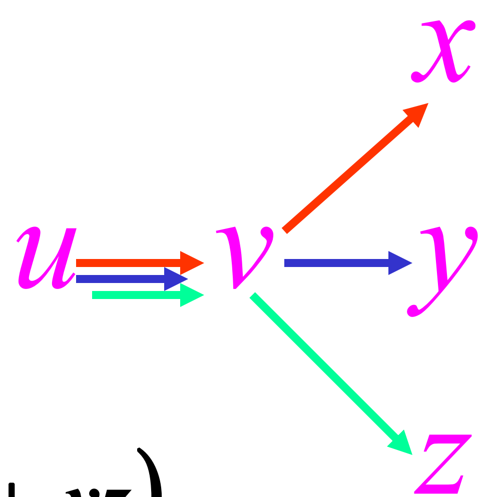
(3)分清是偏导还是导数的 记号!

例5 (1) $u=f(x+xy+xyz)$ 求 $\dfrac{\partial u}{\partial x},\dfrac{\partial u}{\partial y},\dfrac{\partial u}{\partial z}$

(2) $u=f(x,xy,xyz)$ 求 $\dfrac{\partial u}{\partial x},\dfrac{\partial u}{\partial y},\dfrac{\partial u}{\partial z},du$

解 (1) 令 $v=x+xy+xyz$

$$\frac{du}{dv}=f'(v)\qquad \frac{\partial v}{\partial x}=1+y+yz$$



$$\therefore \frac{\partial u}{\partial x}=f'(x+xy+xyz)\cdot(1+y+yz)$$

例5 (1) $u=f\left(x+xy+xyz\right)$ 求 $\dfrac{\partial u}{\partial x},\dfrac{\partial u}{\partial y},\dfrac{\partial u}{\partial z}$

(2) $u=f\left(x,xy,xyz\right)$ 求 $\dfrac{\partial u}{\partial x},\dfrac{\partial u}{\partial y},\dfrac{\partial u}{\partial z},du$

解 (1)令 $v = x + xy + xyz$

$$\therefore \frac{\partial u}{\partial y}=f'\left(x+xy+xyz\right)\cdot\left(x+xz\right)$$

$$\therefore \frac{\partial u}{\partial z}=f'\left(x+xy+xyz\right)\cdot\left(xy\right)$$

$$\begin{tikzcd} & & x \\ u \arrow[r] & v \arrow[ur] \arrow[r] \arrow[dr] & y \\ & & z \end{tikzcd}$$

例5 (2) $u=f(x,xy,xyz)$ 求 $\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z},du$

(2) 令 $\xi=xy,\eta=xyz,u=f(x,\xi,\eta)$

$$\frac{\partial u}{\partial x}=f_1'+f_2'\frac{\partial \xi}{\partial x}+f_3'\frac{\partial \eta}{\partial x}=f_1'+yf_2'+yzf_3'$$

$$\frac{\partial u}{\partial y}=f_2'\frac{\partial \xi}{\partial y}+f_3'\frac{\partial \eta}{\partial y}=f_2'x+f_3'xz$$

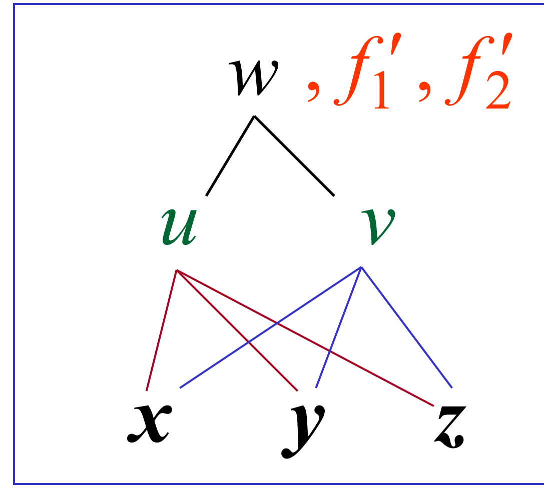
$$\frac{\partial u}{\partial z}=f_3'\frac{\partial \eta}{\partial z}=f_3'xy$$

$$du=\frac{\partial u}{\partial x}dx+\frac{\partial u}{\partial y}dy+\frac{\partial u}{\partial z}dz$$

$$=(f_1'+yf_2'+yzf_3')dx+(f_2'x+f_3'xz)dy+(f_3'xy)dz$$

例6 设 $w=f(x+y+z,xyz)$, f 具有二阶连续偏导数,

求 $\dfrac{\partial w}{\partial x},\dfrac{\partial^2 w}{\partial x\partial z}.$



解 令 $u=x+y+z\,,v=xyz$, 则

$$w=f(u,v)$$

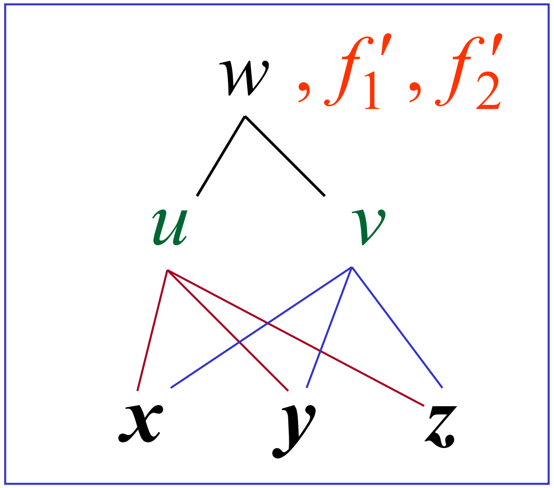
$$\frac{\partial w}{\partial x}=f_1'\cdot 1+f_2'\cdot yz$$

$$=f_1'(x+y+z,xyz)+\underline{yz\,f_2'(x+y+z,xyz)}$$

$$\frac{\partial^2 w}{\partial x\partial z}=f_{11}''\cdot 1+f_{12}''\cdot xy+y\,f_2'+yz[\,f_{21}''\cdot 1+f_{22}''\cdot xy]$$

$$=f_{11}''+y(x+z)f_{12}''+xy^2z\,f_{22}''+y\,f_2'$$

注　二元函数的偏导函数还是二元函数，类似地，三元函数的偏导函数还是三元函数



例6中，$w=f(x+y+z,xyz)$，令$u=x+y+z$，$v=xyz$，

则$w=f(u,v)$，从而$\dfrac{\partial f}{\partial u}=f_1'(u,v)$，$\dfrac{\partial f}{\partial v}=f_2'(u,v)$

条件中f具有二阶连续偏导数，意味着它的混合偏导函数相等

即 $$f_{12}''=\frac{\partial^2 f}{\partial u\,\partial v}=\frac{\partial^2 f}{\partial v\,\partial u}=f_{21}''.$$

练习 设$z=f\left(x^2-y^2,xy\right)$求$\frac{\partial^2 z}{\partial x^2}$、$\frac{\partial^2 z}{\partial x\partial y}$

其中$f\left(u,v\right)$的二阶偏导数连续.

解 $u=x^2-y^2, v=xy$

$$\frac{\partial^2 z}{\partial x^2}=2f_1'+4x^2 f_{11}''+4xyf_{12}''+y^2 f_{22}''$$

$$\frac{\partial^2 z}{\partial x\partial y}=f_2'-4xyf_{11}''+2\left(x^2-y^2\right)f_{12}''+xyf_{22}''$$

例7 设 $z = x^3 f(xy, \frac{y}{x})$,($f$ 具有二阶连续偏导数),

求 $\frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x \partial y}$.

解 用一元函数求导的乘法法则

$$\frac{\partial z}{\partial y} = x^3(f_1' x + f_2' \frac{1}{x}) = x^4 f_1' + x^2 f_2',$$

$$\frac{\partial^2 z}{\partial y^2} = x^4(f_{11}'' x + f_{12}'' \frac{1}{x}) + x^2(f_{21}'' x + f_{22}'' \frac{1}{x})$$

$$= x^5 f_{11}'' + 2x^3 f_{12}'' + x f_{22}'',$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial x}(x^4 f_1' + x^2 f_2')$$

$$= 4x^3 f_1' + x^4[f_{11}'' y + f_{12}''(-\frac{y}{x^2})] + 2xf_2'$$

$$+ x^2[f_{21}'' y + f_{22}''(-\frac{y}{x^2})]$$

$$= 4x^3 f_1' + 2xf_2' + x^4 yf_{11}'' - yf_{22}''.$$

练习 已知 $z = x^2 y\, f\left(x^2 - y^2, xy\right)$ 求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

解 用求导的乘法法则

$$\frac{\partial z}{\partial x} = 2xyf\left(x^2 - y^2, xy\right) + x^2 y\left[2xf_1' + yf_2'\right]$$

$$\frac{\partial z}{\partial y} = x^2\, f\left(x^2 - y^2, xy\right) + x^2 y\left[-2yf_1' + xf_2'\right]$$

练习

1. 已知 $f(x,x^2)=1$, $f_1'(x,x^2)=2x$, 求 $f_2'(x,x^2)$.

2. 设函数 $z=f(x,y)$ 在点(1,1)处可微，且

$$f(1,1)=1,\quad \left.\frac{\partial f}{\partial x}\right|_{(1,1)}=2,\quad \left.\frac{\partial f}{\partial y}\right|_{(1,1)}=3,$$

$\varphi(x)=f(x,f(x,x))$, 求 $\left.\frac{\mathrm{d}}{\mathrm{d}x}\varphi^3(x)\right|_{x=1}$. **(2001考研)**

解答：

1. 已知 $f(x,x^2)=1$，$f_1'(x,x^2)=2x$，求 $f_2'(x,x^2)$.

解 由 $f(x,x^2)=1$ 两边对 x 求导, 得

$$f_1'(x,x^2)+f_2'(x,x^2)\cdot 2x=0$$

$$f_1'(x,x^2)=2x$$

$$f_2'(x,x^2)=-1$$

2. 设函数 $z=f(x,y)$ 在点 $(1,1)$ 处可微，且

$$f(1,1)=1,\quad \left.\frac{\partial f}{\partial x}\right|_{(1,1)}=2,\quad \left.\frac{\partial f}{\partial y}\right|_{(1,1)}=3,$$

$\varphi(x)=f(x,f(x,x))$，求 $\left.\dfrac{\mathrm{d}}{\mathrm{d}x}\varphi^3(x)\right|_{x=1}$. **(2001考研)**

解: 由题设 $\varphi(1)=f(1,f(1,1))=f(1,1)=1$

$$\left.\frac{\mathrm{d}}{\mathrm{d}x}\varphi^3(x)\right|_{x=1}=\left.3\varphi^2(x)\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_{x=1}$$

$$=3\Big[f_1'(x,f(x,x))$$

$$\left.+f_2'(x,f(x,x))\big(f_1'(x,x)+f_2'(x,x)\big)\Big]\right|_{x=1}$$

$$=3\cdot\big[2+3\cdot(2+3)\big]=51$$

二、全微分形式不变性

设函数$z=f(u,v)$具有连续偏导数，则有全微分

$$dz=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv;$$当$u=\phi(x,y)$、$v=\psi(x,y)$

时，有$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy.$

全微分形式不变性的实质：

无论 z是自变量u、v的函数或中间变量u、v的函数，它的全微分形式是一样的.

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

$$= \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}\right)dx + \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}\right)dy$$

$$= \frac{\partial z}{\partial u}\left(\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy\right) + \frac{\partial z}{\partial v}\left(\frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy\right)$$

$$= \frac{\partial z}{\partial u}du + \frac{\partial z}{\partial v}dv.$$

补充：多元函数全微分的四则运算法则

$$d\left(z_1 \pm z_2\right) = d\left(z_1\right) \pm d\left(z_2\right)$$

$$d\left(z_1 z_2\right) = z_1 d\left(z_2\right) + z_2 d\left(z_1\right)$$

$$d\left(\frac{z_1}{z_2}\right) = \frac{z_2 d\left(z_1\right) - z_1 d\left(z_2\right)}{{z_2}^2}$$

例 8 已知$e^{-xy}-2z+e^{z}=0$，求$\frac{\partial z}{\partial x}$和$\frac{\partial z}{\partial y}$.

解 $$dz=\frac{\partial z}{\partial x}\Delta x+\frac{\partial z}{\partial y}\Delta y=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy$$

$$\because d(e^{-xy}-2z+e^{z})=0,$$

$$\therefore e^{-xy}d(-xy)-2dz+e^{z}dz=0,$$

$$(e^{z}-2)dz=e^{-xy}(xdy+ydx)$$

$$dz=\frac{ye^{-xy}}{(e^{z}-2)}dx+\frac{xe^{-xy}}{(e^{z}-2)}dy$$

$$\frac{\partial z}{\partial x}=\frac{ye^{-xy}}{e^{z}-2},\qquad \frac{\partial z}{\partial y}=\frac{xe^{-xy}}{e^{z}-2}.$$

三、变量代换(偏微分方程的恒等变形)

定义 称含有未知函数及其偏导数的方程为偏微分方程。满足偏微分方程的多元函数称为偏微分方程的解.

例：$y\dfrac{\partial z}{\partial x}+x\dfrac{\partial z}{\partial y}=0,\quad \dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=1$

例9 在自变量变换$u=x, v=x^2-y^2$下，求方程

$$y\frac{\partial z}{\partial x}+x\frac{\partial z}{\partial y}=0$$的解。

解 视x, y为自变量，u, v为中间变量。

$$\frac{\partial z}{\partial x}=f_1'+f_2'\cdot 2x,\quad \frac{\partial z}{\partial y}=-2y\cdot f_2'$$

$$y\frac{\partial z}{\partial x}+x\frac{\partial z}{\partial y}= f_1'\cdot y+f_2'\cdot 2xy-2xyf_2'$$

$$\Rightarrow \frac{\partial z}{\partial u}=0,\qquad \therefore z=f(v)=f(x^2-y^2).$$

复习：

推论 如果对任意的$x\in(a, b)$都有$f'(x)=0$, 则$f(x)$在(a, b)内恒为一常数 .

证明：设x_1,x_2是(a, b)内任意两点，由拉格朗日定理有

$$f(x_2)-f(x_1)=f'(\xi)(x_2-x_1)=0 \quad (\xi\text{在}x_1,x_2\text{之间})$$

$$\therefore f(x_2)=f(x_1)$$

由x_1, x_2的任意性知: $f(x)$=常数, $x\in(a, b)$.

例10 设 $u=f(x,y)$ 二阶偏导数连续，求下列表达式在极坐标系下的形式：$(\frac{\partial u}{\partial x})^2+(\frac{\partial u}{\partial y})^2$

解: 已知 $x=r\cos\theta$，$y=r\sin\theta$, 则

$$r=\sqrt{x^2+y^2}\ ,\ \theta=\arctan\frac{y}{x}$$

$$\frac{\partial u}{\partial x}=\frac{\partial u}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial x}$$

$$\boxed{\frac{\partial r}{\partial x}=\frac{x}{r},\quad \frac{\partial \theta}{\partial x}=\frac{\frac{-y}{x^2}}{1+(\frac{y}{x})^2}=\frac{-y}{x^2+y^2}}$$

$$=\frac{\partial u}{\partial r}\frac{x}{r}-\frac{\partial u}{\partial \theta}\frac{y}{r^2}=\frac{\partial u}{\partial r}\cos\theta-\frac{\partial u}{\partial \theta}\frac{\sin\theta}{r}$$

$$\frac{\partial u}{\partial y}=\frac{\partial u}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial y}$$

$$\boxed{\frac{\partial r}{\partial y}=\frac{y}{r},\ \frac{\partial \theta}{\partial y}=\frac{\frac{1}{x}}{1+(\frac{y}{x})^2}=\frac{x}{x^2+y^2}}$$

$$=\frac{\partial u}{\partial r}\frac{y}{r}+\frac{\partial u}{\partial \theta}\frac{x}{r^2}$$

$$=\frac{\partial u}{\partial r}\sin\theta+\frac{\partial u}{\partial \theta}\frac{\cos\theta}{r}$$

$$\therefore\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=\left(\frac{\partial u}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$$

例11 设变换 $\begin{cases} u = x - 2y \\ v = x + ay \end{cases}$ 可把方程 $6\dfrac{\partial^2 z}{\partial x^2} + \dfrac{\partial^2 z}{\partial x \partial y} - \dfrac{\partial^2 z}{\partial y^2} = 0$

简化为 $\dfrac{\partial^2 z}{\partial u \partial v} = 0$，求常数$a$。

解 $\dfrac{\partial z}{\partial x} = \dfrac{\partial z}{\partial u} + \dfrac{\partial z}{\partial v}$, $\dfrac{\partial z}{\partial y} = -2\dfrac{\partial z}{\partial u} + a\dfrac{\partial z}{\partial v}$,

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2} + 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}, \quad \frac{\partial^2 z}{\partial y^2} = 4\frac{\partial^2 z}{\partial u^2} - 4a\frac{\partial^2 z}{\partial u \partial v} + a^2\frac{\partial^2 z}{\partial v^2},$$

$$\frac{\partial^2 z}{\partial x \partial y} = -2\frac{\partial^2 z}{\partial u^2} + (a-2)\frac{\partial^2 z}{\partial u \partial v} + a\frac{\partial^2 z}{\partial v^2}.$$

将上述结果代入原方程，经整理后得

$$(10 + 5a)\frac{\partial^2 z}{\partial u \partial v} + (6 + a - a^2)\frac{\partial^2 z}{\partial v^2} = 0.$$

依题意a应满足 $6 + a - a^2 = 0$ 且 $10 + 5a \neq 0$ 解得 $a=3$。

四、小结

1、链式法则

2、全微分形式不变性

（理解其实质）